

Subzero.jl: A Coupled, Julia Version of a Discrete-Element Sea Ice Model

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Summary

Subzero.jl is a discrete-element model (DEM) for simulating the dynamics of two-dimensional polygonal sea-ice floes. The model is a port of the MATLAB-based SubZero model (Montemuro & Manucharyan, 2025) to the Julia programming language (Bezanson et al., 2017), with substantial re-engineering to increase performance, improve extensibility and usability, and provide native Julia infrastructure for two-way coupling with the external ocean dynamics model Oceananigans.jl (Ramadhan et al., 2020; Wagner et al., 2025). Subzero.jl is available as a registered Julia package.

Statement of need

Arctic sea ice extent and concentration have declined at rates that are commonly underestimated by climate projection models (Notz & Community, S.I.M.I.P., 2020; Shu et al., 2020). Potential sources of uncertainty arise from an inaccurate representation of interactions between the ocean and sea ice within these climate models, as well as the assumption of continuum sea-ice dynamics for the sake of reducing computational complexity.

Discrete-element models (DEMs), where each piece of sea ice is represented as an individual simulation element, all of which can dynamically interact, are used to explore these uncertainties and study fine-scale sea ice dynamics. While a range of sea ice DEMs are available (Damsgaard et al., 2018; Herman, 2013, 2016; Hopkins, 2004; Rabatel et al., 2015), many of these models make a range of simplifications, particularly in the geometric representation of the ice floes.

The MATLAB-based SubZero model (Montemuro & Manucharyan, 2025) allows for complex and evolving floe shapes, but is computationally expensive. Moreover, due to the highly connected nature of sea ice floes and ocean processes (Brenner et al., 2023; Gupta et al., 2024; Gupta & Thompson, 2022; Horvat & Tziperman, 2018), there is a need for DEM simulations to be coupled to a dynamic ocean model to explore two-way feedbacks, which is a feature not readily available for most extant models.

We present Subzero.jl, a native Julia (Bezanson et al., 2017) version of the MATLAB discrete-element model SubZero (Manucharyan & Montemuro, 2022; Montemuro & Manucharyan, 2025) that addresses both of these problems.

Functionality

Subzero.jl represents sea ice floes as polygonal elements that move in response to forcing from the atmosphere and the ocean (Manucharyan & Montemuro, 2022). Over time, a given floe may change its horizontal shape, adjust vertical thickness, and fracture into multiple pieces

39 as a result of interactions with other floes and topographical elements. Subzero.jl improves
40 upon the MATLAB version in three major ways: (i) a modular interface that reduces the need
41 for users to modify source code, (ii) the ability to couple to a performant native-Julia ocean
42 model, and (iii) significant enhancement in computational speed.

43 **Modular interface**

44 This new Julia implementation provides a more extensible, user-friendly interface while preserving
45 the scientific behavior of the previous MATLAB model. In the previous MATLAB workflow,
46 simulation-specific settings were managed at the repository level, requiring copying and
47 adapting directories to create new experiments. However, Subzero.jl is organized around
48 composable object types (structs) in a hierarchical structure, with a top-level Simulation
49 object containing Floes, Domain, Atmospheric Forcing, Ocean Forcing, Physical Settings
50 and Output Writer components. Using dedicated run scripts, the user may modify object
51 parameters to easily turn on/off physical processes such as fracturing, ridging, and welding.
52 Moreover, the use of Julia's multiple dispatch allows flexibility to add more functionality
53 without changing the underlying model structure. This flexibility is highlighted in the tutorial
54 and examples provided with the documentation. Additionally, Subzero.jl includes a suite of
55 unit and integration tests that verify consistency with the original MATLAB model and ensure
56 conservation of energy and momentum. These changes allow for more dynamic and flexible
57 usage and development than in the MATLAB version, while ensuring parity with the original
58 code.

59 **Ice-Ocean Coupling**

60 The code now also includes a framework that enables two-way mechanically coupled simulations
61 with the Julia ocean model Oceananigans.jl (Ramadhan et al., 2020; Wagner et al., 2025).
62 In this configuration, sea-ice floes in Subzero.jl are forced by gridded ocean velocity fields
63 supplied by Oceananigans.jl and interpolated onto a mesh carried by each floe. In turn,
64 floe-resolved ice-ocean stresses computed by Subzero.jl are interpolated onto the ocean grid
65 and returned to the ocean model as a spatially heterogeneous and temporally evolving surface
66 boundary condition. The exchange of fields is implemented using a callback functionality in
67 Oceananigans.jl, allowing serial coupling at user-defined frequencies. This coupled DEM-LES
68 system resolves the mechanical interaction between evolving ocean fields and individual sea ice
69 floes, enabling investigation of floe-scale modulation of upper-ocean dynamics (Brenner et al.,
70 2025). An example coupled simulation is demonstrated in Figure 1.

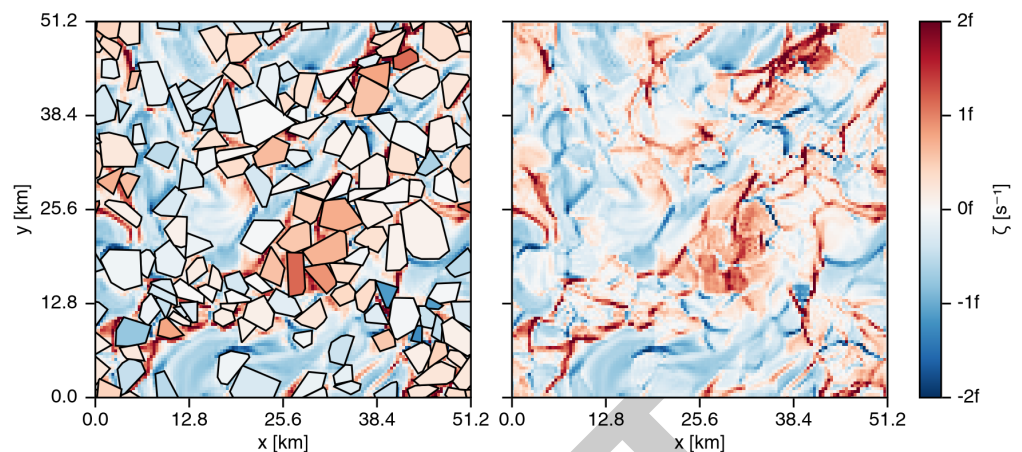


Figure 1: Example Subzero-Oceananigans coupled simulation. Left: surface fields of ocean vorticity, overlain by sea ice floes, each coloured by their vorticity. Right: the same ocean vorticity field with floes not plotted; the impacts of ice-ocean coupling are still evident in the "patchiness" of the field. Vorticity units are given in terms of the Coriolis frequency, f , which has a value of $1.26 \times 10^{-4} \text{ s}^{-1}$ at 60°N .

Performance improvements

Beyond this new functionality, Subzero.jl provides significant performance improvements relative to the original MATLAB implementation of the model. We used the `shear_flow` example from the [Subzero.jl repository](#) as a basis for testing speed enhancements. The simulation includes 50 floes at 75% sea ice concentration, forced by sheared ocean currents in a doubly-periodic domain; we also created a version with 1000 floes, and then implemented both versions in the MATLAB SubZero codebase. We tested the models on the California Institute of Technology's "Resnick" High Performance Computing system, on a single compute node using 1-16 CPU cores, with each configuration repeated five times. Results in [Figure 2](#) show comparisons of both end-to-end runtime (including initialization and I/O) and simulation time (time spent advancing the model state).

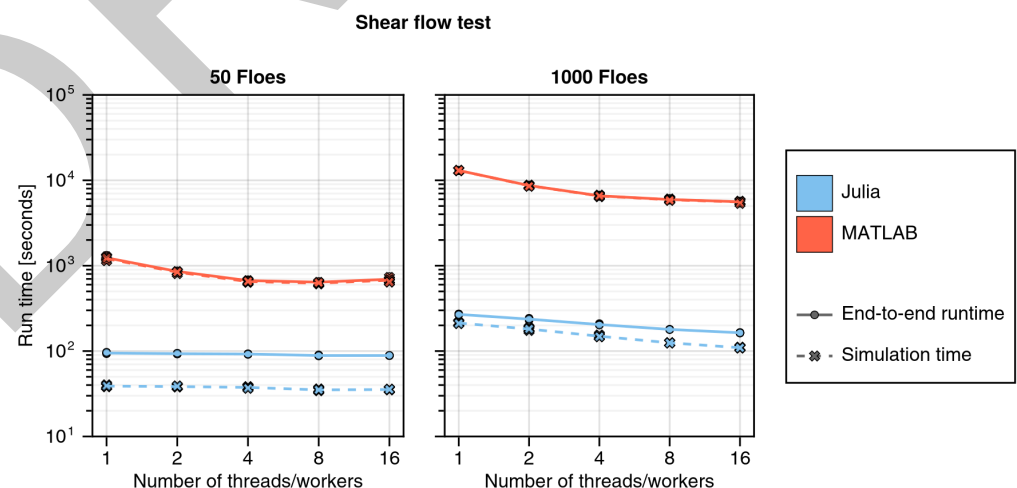


Figure 2: Runtime of `shear_flow` simulation in Julia Subzero.jl (blue) and MATLAB SubZero (red) for comparison. Left: 50 floe simulation. Right: 1000 floe simulation. Solid lines show end-to-end time while dashed lines show strictly simulation time.

Subzero.jl achieves substantial speedups relative to the MATLAB implementation: 7.2-13.0

times faster (end-to-end) and 17.4–31.1 times faster (simulation) for 50 floes, increasing to 32.2–48.5 times (end-to-end) and 43.9–61.1 times (simulation) for 1000 floes. Parallel scaling is modest in both implementations, with simulation time speedups of 1.1 times (Julia) and 1.8 times (MATLAB) for 50 floes, and 1.9 times (Julia) and 2.3 times (MATLAB) for 1000 floes when increasing from 1 to 16 cores. The speed increase of Subzero.jl relative to the original MATLAB model will enable longer and more complex simulations, particularly at higher floe counts.

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